I-SOLEY Validation Against Real-World PV Production Data

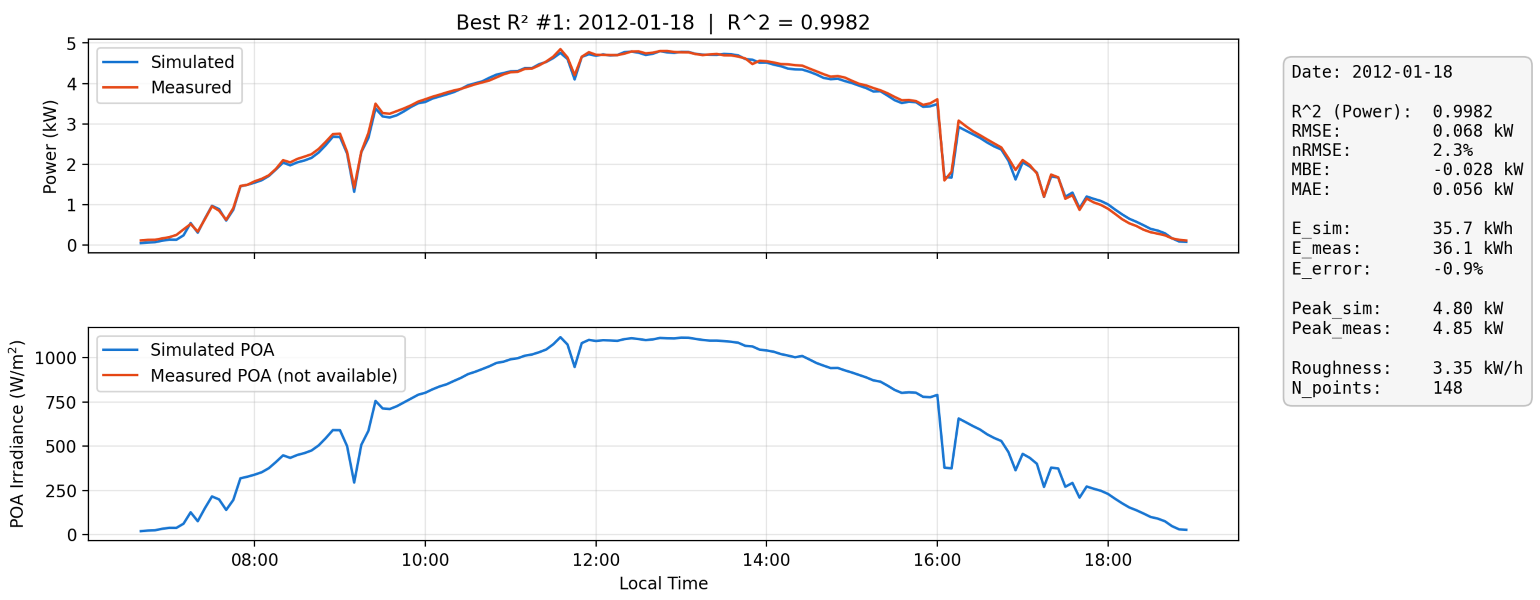
DKASC Alice Springs, Australia | 5-min resolution | Two technologies | Full-year comparison

SOLEY's outdoor module (I-SOLEY) was validated against ground-truth production data from the Desert Knowledge Australia Solar Centre (DKASC). Two real-world systems, "M2C-2012" (SunPower SPR-238 mono-Si, 5.24 kWp) and M18A-2015 (First Solar FS-387 thin-film CdTe, 5.1 kWp), were simulated using the on-site ground meteorological measurements (GHI, DHI, wind, temperature) provided by DKASC and manufacturer datasheets only, with no post-hoc fitting to measured electrical output. Satellite irradiance databases (ERA5, SARAH-2/3, NSRDB) are also supported but were not used here, as ground data allows isolating the accuracy of the electrical model itself. The simulation resolves power at 5-minute intervals across a full calendar year.

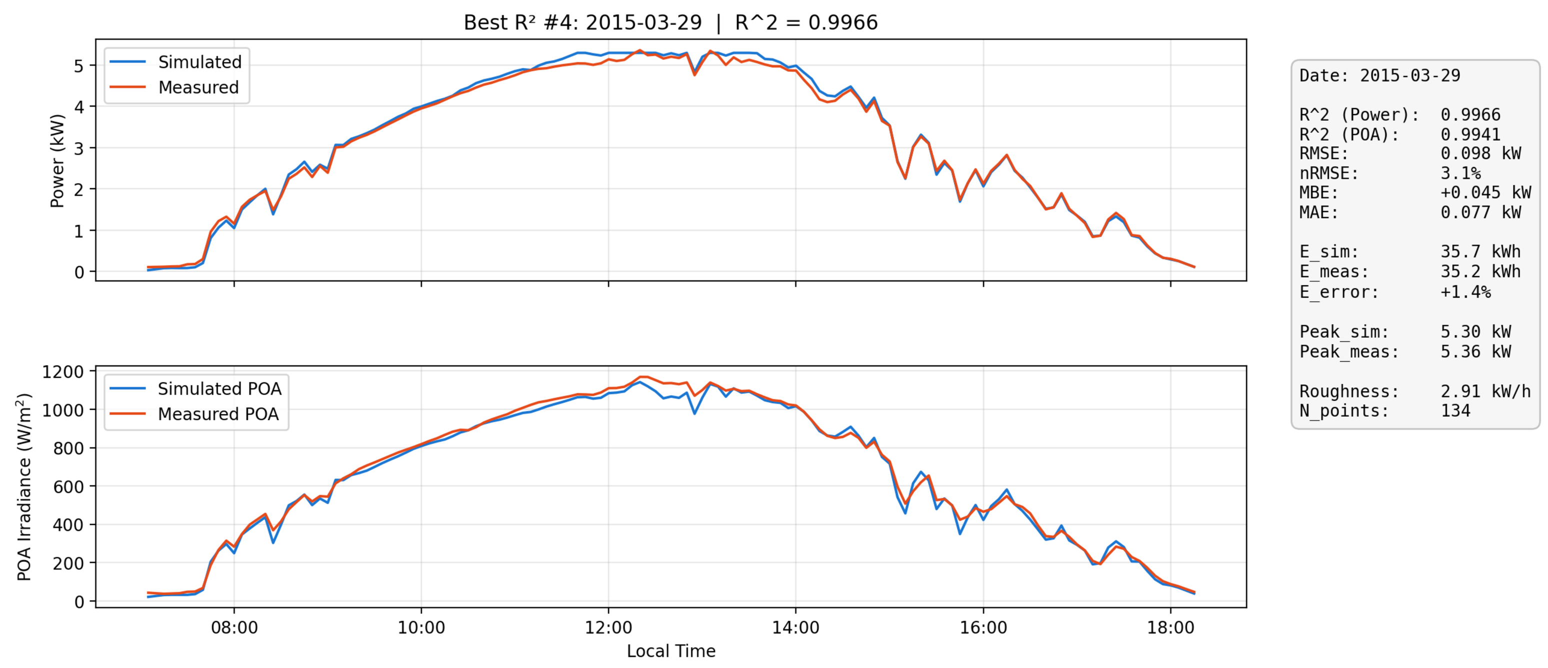
$R^2 = 0.97$	< 1 % annual energy error	100 000+	MBE ≈ 0 kW
annual (instantaneous)	best-tuned config	data points / system	near-zero bias

Clear-Sky Conditions — $R^2 > 0.99$

SunPower SPR-238 (mono-Si, 5.24 kWp)

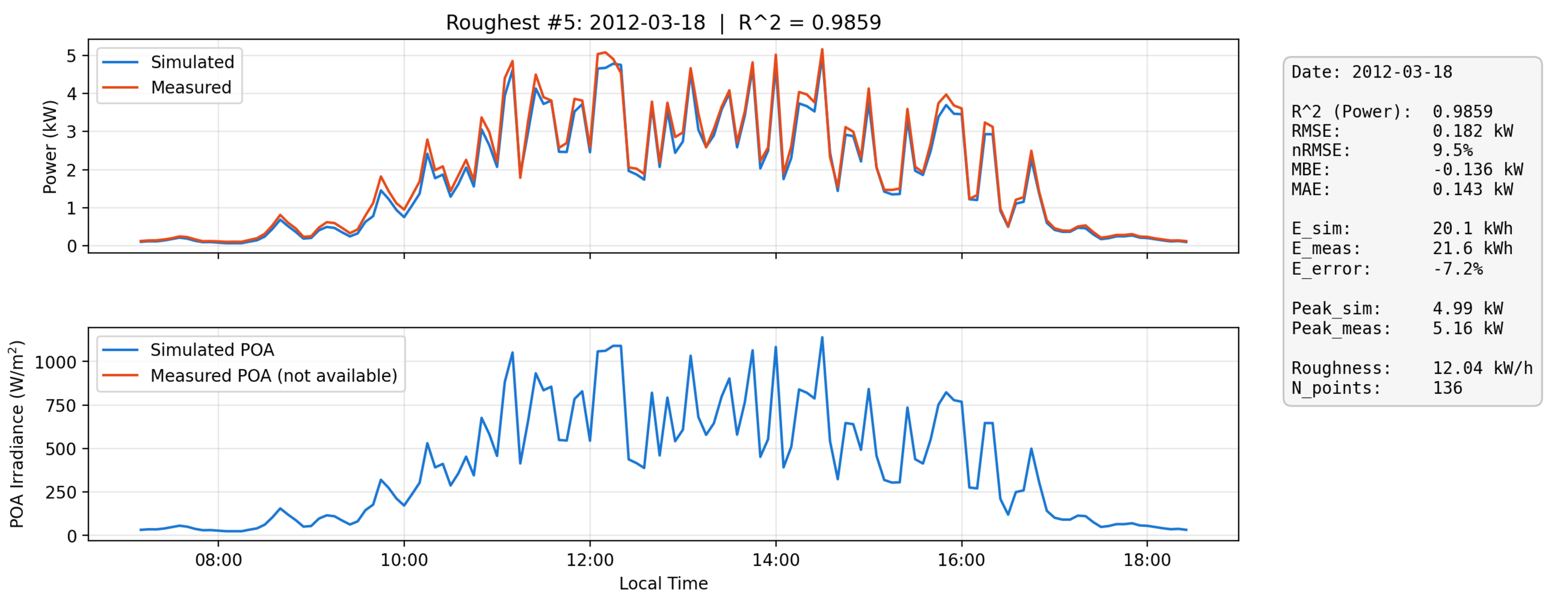


First Solar FS-387 (thin-film CdTe, 5.1 kWp)

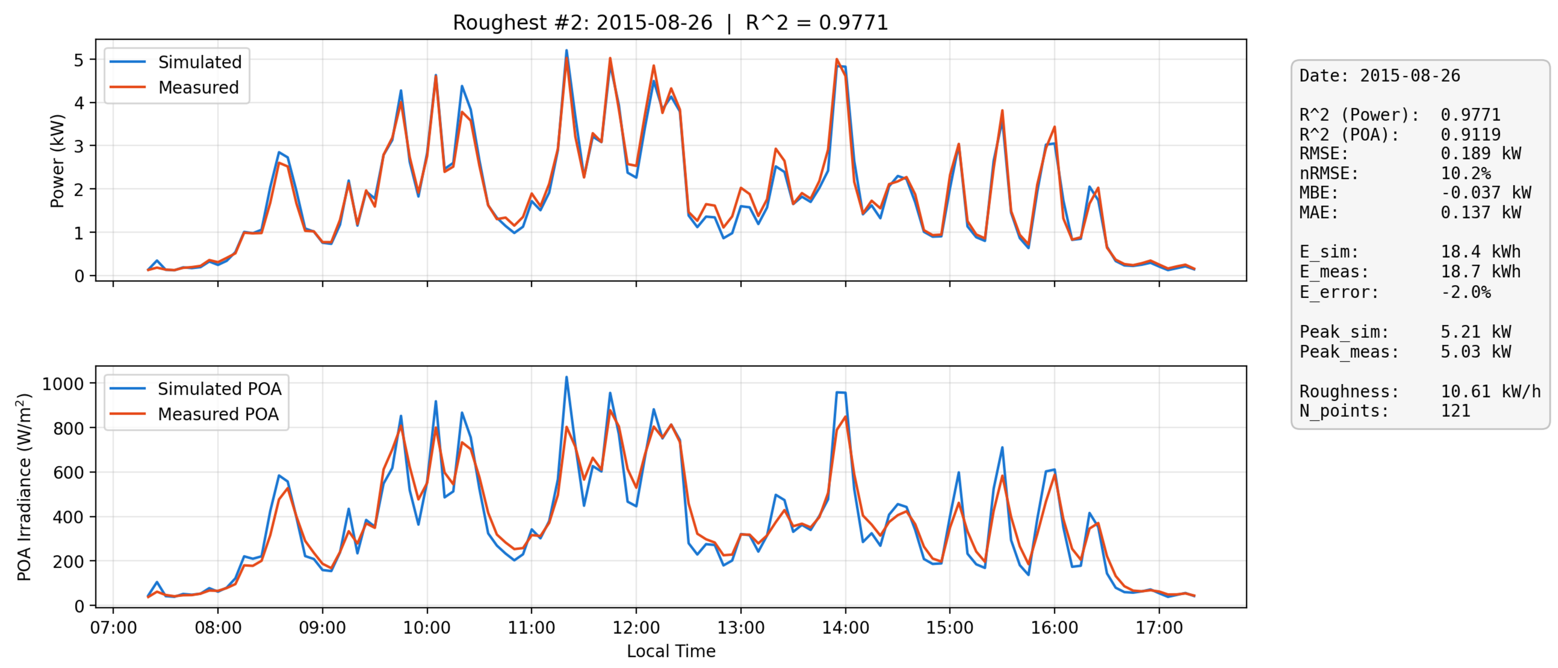


Variable / Cloudy Conditions — $R^2 > 0.97$ under rapidly varying irradiance

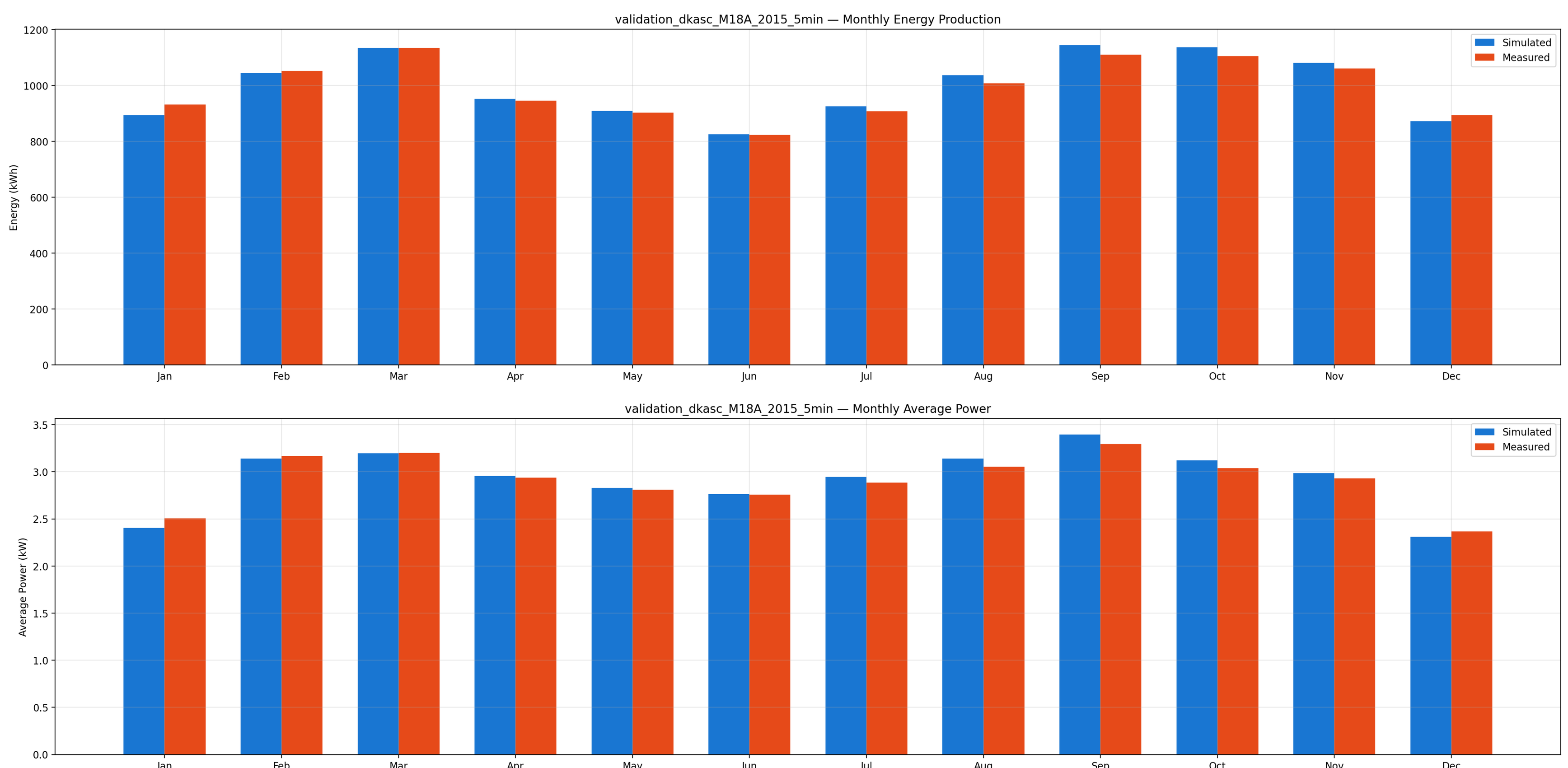
SPR-238 — cloudy ($R^2 = 0.986$)

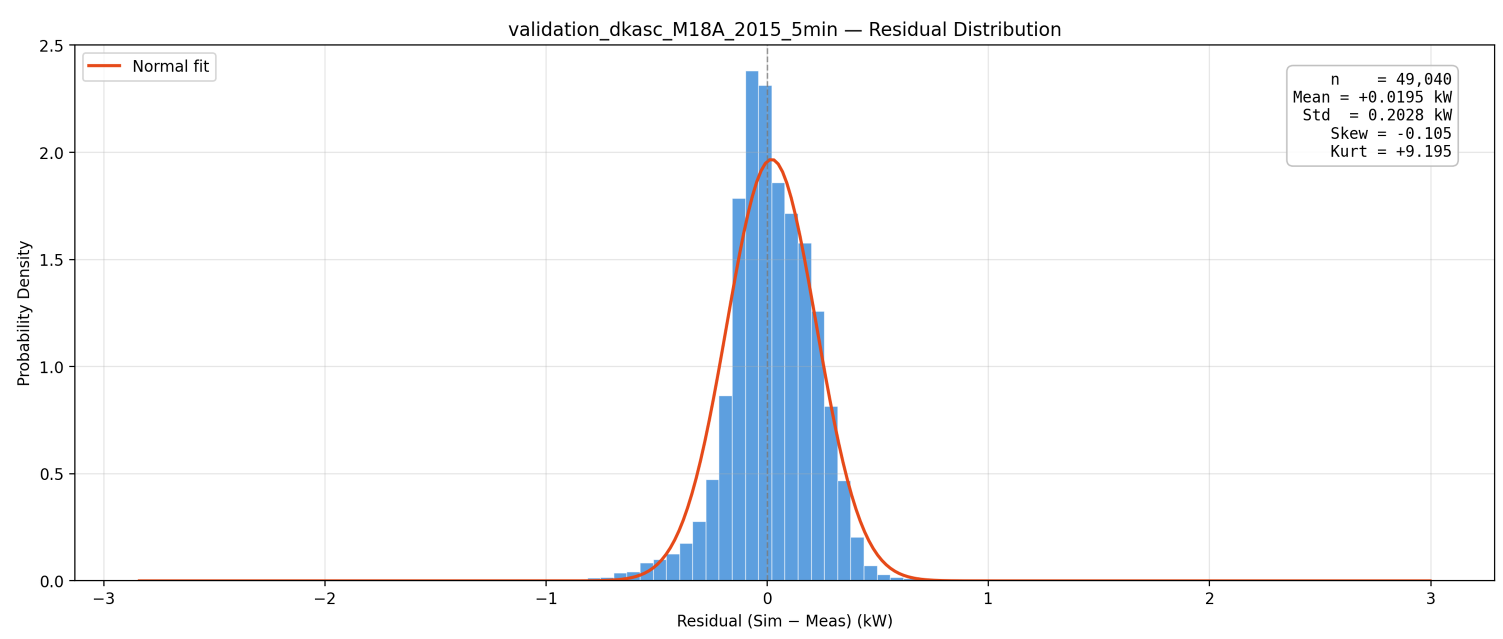


FS-387 — cloudy ($R^2 = 0.977$)

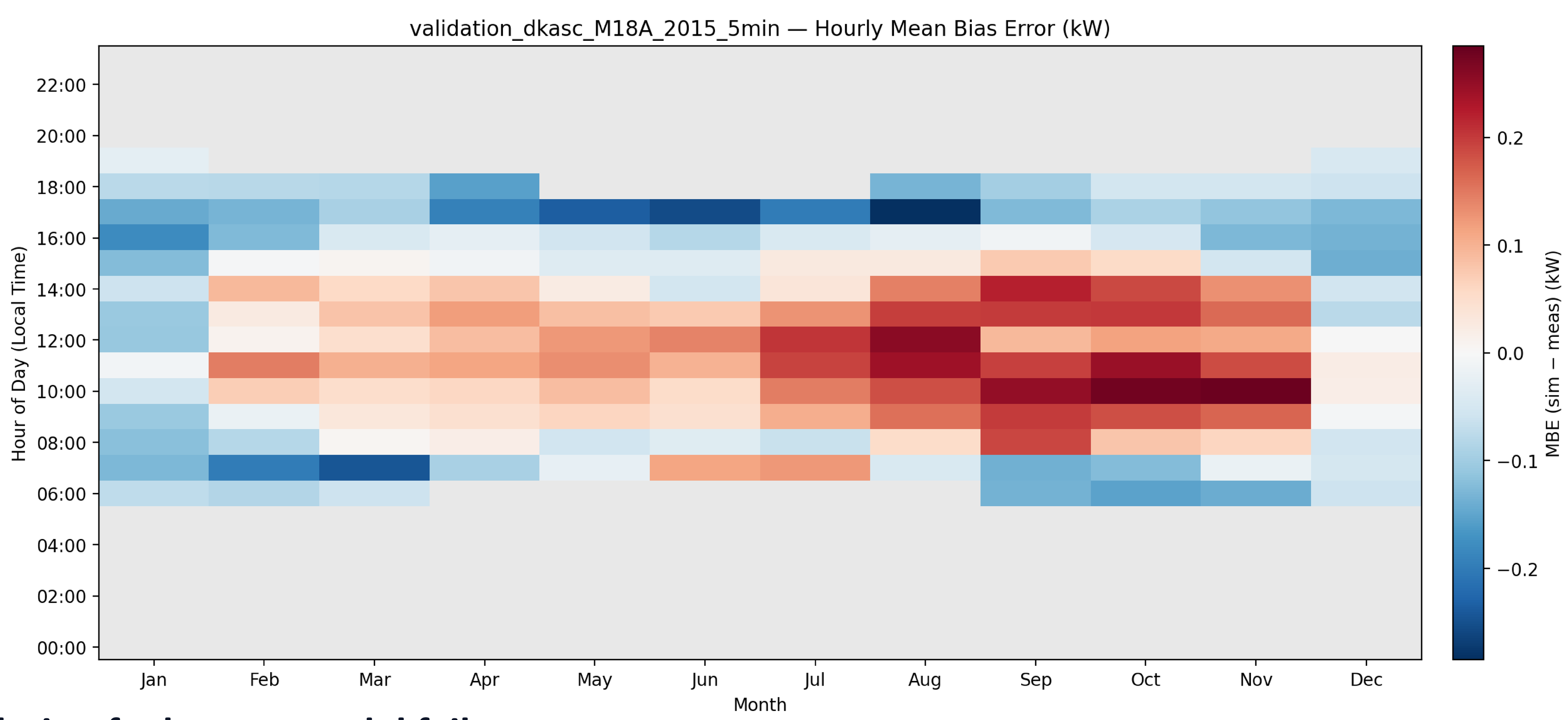


Year-Round Metrics (FS-387, 2015) — Monthly accuracy & near-zero systematic bias

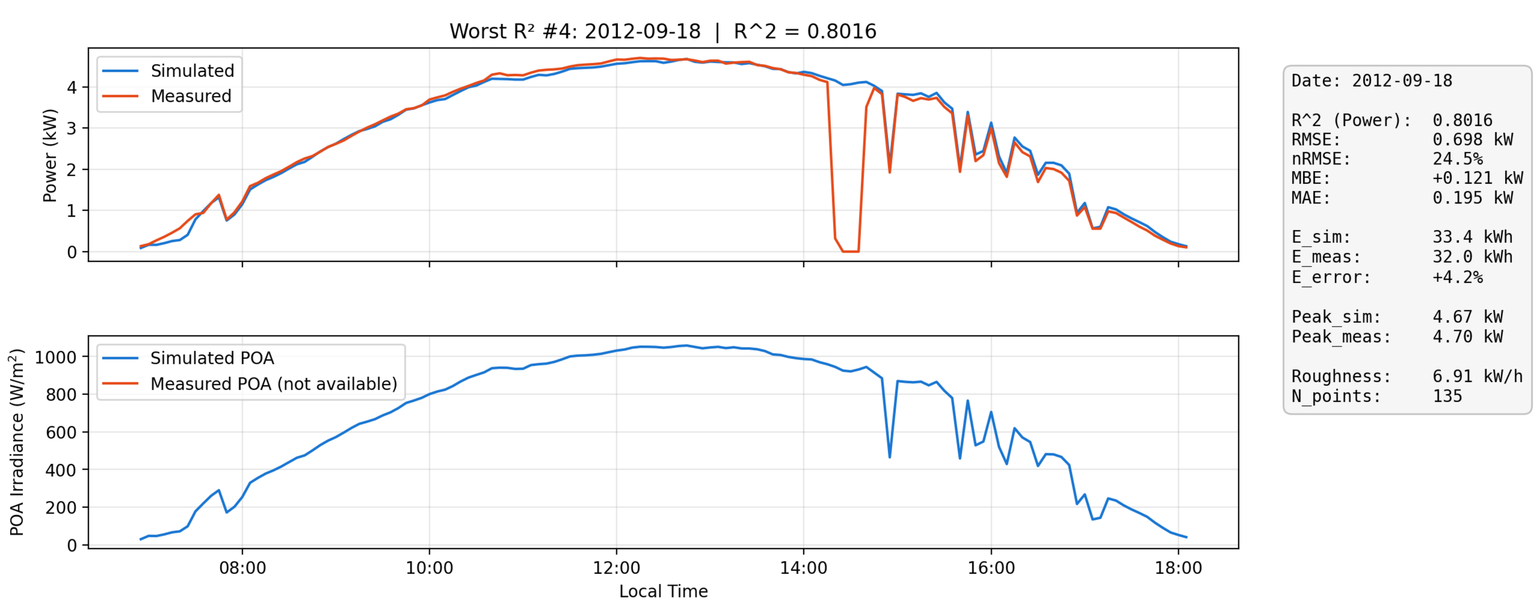




Hourly Mean Bias Error — Seasonal pattern from installation-side effects, not model drift



Interpreting Discrepancies — Installation faults, not model failures



One of the worst-performing day in the SPR-238 dataset ($R^2 = 0.80$). The sharp power drops in the measured data are characteristic of inverter clipping or string disconnection which are installation-specific faults, not physical phenomena.

The simulated curve follows a physically correct trajectory: the model is right; the installation is misbehaving. This pattern is consistent across all large-residual days.

With I-SOLEY, we can inject both discrete, event-driven faults, such as string disconnections, glass breakage, and sudden soiling drops, as well as progressive degradation modes like Potential Induced Degradation (PID) and solder fatigue. Furthermore, these injected faults can be programmed with highly specific temporal distributions, including random, seasonal, or continuous recurrences with customizable recovery times, allowing for the generation of perfectly labelled, highly realistic training datasets.

Validated predictions with < 1% annual bias and $R^2 > 0.97$ enable:

- LCA: Auditable, physics-based energy-yield data for life-cycle energy and carbon accounting of PV systems.
- Real-time fault detection: The residual (simulated - measured) acts as a live diagnostic signal, flagging inverter trips, string faults, and shading anomalies as they occur.
- Synthetic data factory for ML: Generate virtually unlimited high-quality production data for any location, orientation, and technology, enabling ML training without costly measurement campaigns.

Data: DKASC (https://dkasolarcentre.com.au/) | Meteorology: on-site ground measurements | Simulation: Zacharie JEHL LI-KAO with SOLEY-SUITE | Contact: zacharie.jehl@upc.edu

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